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LOAD CONTROL ARCHITECTURE OF AN ENERGY CONTROL SYSTEM

Abstract

The present disclosure provides an electrical system that includes an energy control system, a photovoltaic (PV) power generation system electrically coupled to the energy control system, an energy storage system electrically coupled to the energy control system, and a smart load panel electrically coupled to the energy control system and to a plurality of backup loads. The energy control system operates in an on-grid mode electrically connecting the PV power generation system to a utility grid and a backup mode electrically disconnecting the PV power generation system from the utility grid. The smart load panel selectively disconnects one or more of the plurality of backup loads from the energy control system when the energy control system is in the on-grid mode and when the energy control system is in the backup mode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. application Ser. No. 18/352,596, filed Jul. 14, 2023, which is a continuation of U.S. application Ser. No. 17/731,817, filed Apr. 28, 2022, now issued as U.S. Pat. No. 11,705,736, issued on Jul. 18, 2023, which claims priority to U.S. Provisional Patent Application No. 63/181,524 filed on Apr. 29, 2021, which are all incorporated by reference herein in their entirety for all purposes.

FIELD

[0002] The present disclosure relates to systems and methods for integrating a plurality of electrical loads to a backup side of an energy control system.

BACKGROUND

[0003] Existing backup power supply systems, such as PV systems, for commercial buildings or residential homes sometimes include storage systems (e.g., a combination of batteries and an inverter) to store power when PV power output exceeds load demand and to provide power when PV power output cannot match load demand during microgrid operation. However, larger loads of local electrical systems, such as air conditioners, electric vehicle chargers, pool pumps, range ovens, etc., are usually not wired to the backup side of the electrical system because large loads tend to drain the storage system rapidly or overload the storage system during microgrid operation. Furthermore, National Electrical Code requires any electrical loads that exceed the capacity of the local energy storage system during microgrid mode to be wired to the non-backup side of the local electrical system. While a majority of residential loads are usually smaller (e.g., less than 40 A) and can be easily configured to receive power supply from the local energy storage system, implementing larger loads (e.g., 40 A or greater) to the backup side of the electrical system is typically tedious and expensive.

[0004] Thus, integrating all loads of a local electrical system to a backup side of an energy control system can be challenging.

BRIEF SUMMARY

[0005] Accordingly, there is a need for systems and procedures to allow large loads (e.g., 40 A or greater) to be connected to the backup side of an electrical system while providing a user the capability to selectively turn off any load, including the large loads, during backup mode.

Disposing all or most electrical loads on the backup side of the electrical system, while still maintaining control of the electrical system, has numerous advantages, such as minimizing grid power usage during on-grid mode and allowing the user to selectively provide backup power to each load of the local electrical system.

[0006] In some embodiments, the present disclosure provides an electrical system. In some embodiments, the electrical system includes an energy control system. In some embodiments, the electrical system includes a PV power generation system electrically coupled to the energy control system. In some embodiments, the PV power generation system is configured to generate and supply power. In some embodiments, the electrical system includes an energy storage system electrically coupled to the energy control system. In some embodiments, the energy storage system is configured to store power supplied by the PV power generation system and discharge stored

power to the energy control system. In some embodiments, the electrical system includes a smart load panel electrically coupled to the energy control system and to a plurality of backup loads. In some embodiments, the energy control system is configured to operate in an on-grid mode electrically connecting the PV power generation system to a utility grid and a backup mode electrically disconnecting the PV power generation system from the utility grid. In some embodiments, the smart load panel is configured to selectively disconnect one or more of the plurality of backup loads from the energy control system when the energy control system is in the on-grid mode and when the energy control system is in the backup mode. In some embodiments, the smart load panel is configured to selectively disconnect a majority of the plurality of backup loads from the energy control system when the energy control system is in the on-grid mode and when the energy control system is in the backup mode. In some embodiments, the smart load panel is configured to selectively disconnect each of the plurality of backup loads from the energy control system when the energy control system is in the on-grid mode and when the energy control system is in the backup mode.

[0007] In some embodiments, the smart load panel includes a load panel controller in communication with a user device. In some embodiments, upon receiving a disconnection command from the user device, the load panel controller is configured to electrically disconnect one or more of the plurality of backup loads from the energy control system when the energy control system is in the on-grid mode and when the energy control system is in the backup mode.

[0008] In some embodiments, the smart load panel is configured to maintain electrical connection of the plurality of backup loads to the energy control system when the energy control system is in the on-grid mode and to electrically disconnect one or more of the plurality of backup loads automatically from the energy control system when the energy control system switches from the on-grid mode to the backup mode. In some embodiments, the smart load panel includes a load panel controller configured to detect the frequency or voltage of AC power supplied to the plurality of backup loads. In some embodiments, the load panel controller is configured to determine whether the energy control system has switched from the on-grid mode to the backup mode based on the detected frequency or voltage of AC power supplied to the plurality of backup loads.

[0009] In some embodiments, the energy control system includes a system controller and the smart load panel includes a load panel controller in communication with the system controller. In some embodiments, upon receiving a disconnection command from the system controller, the load panel controller is configured to electrically disconnect the one or more backup loads from the energy control system when the energy control system is in the on-grid mode and when the energy control system is in the backup mode. In some embodiments, the load panel controller is in communication with the system controller according to a wireless communication standard. In some embodiments, the wireless communication standard is a Bluetooth standard. In some embodiments, the wireless communication standard is Wi-Fi. In some embodiments, the wireless communication standard is an AC Power Line Communication. In some embodiments, the load panel controller is in communication with the system controller over a wired connection. In some embodiments, the wired connection is based on at least one of direct logic signal(s), RS-485, RS-232, Controller Area Network, and Modbus protocol.

[0010] In some embodiments, the electrical system includes a sub-panel electrically coupled to the one or more backup loads and the smart load panel. In some embodiments, the smart load panel is located upstream of the sub-panel and downstream of the energy control system. In some embodiments, the smart load panel is located downstream of the sub-panel and upstream of the one or more backup loads. In some embodiments, the smart load panel comprises at least one of an electromechanical breaker, a relay, and a transistor switch. In some embodiments, the one or more backup loads comprises a rating of 40 amps or greater.

[0011] The present disclosure also provides methods for integrating a plurality of electrical loads to a backup side of an energy control system. In some embodiments, the method includes a step of

determining a location of one or more of the plurality of electrical loads with respect to the energy control system. In some embodiments, the method includes a step of determining a load control protocol associated with the one or more electrical loads. In some embodiments, the method includes a step of selecting an electrical disconnection component based on the determined load control protocol of the one or more electrical loads. In some embodiments, the method includes a step of connecting the selected electrical disconnection component to the one or more electrical loads and to the backup side of the energy control system. In some embodiments, the method includes a step of selecting a mode of communication between the selected electrical disconnection component and the energy control system based on the determined location of the one or more electrical loads and the determined load control protocol.

[0012] In some embodiments, the load control protocol includes allowing a user to selectively connect and disconnect the one or more electrical loads on demand. In some embodiments, the load control protocol includes automatically disconnecting the one or more electrical loads from the energy control system based on a detected frequency or voltage of AC power supply. In some embodiments, the electrical disconnection component includes at least one of an electromechanical breaker, a relay, and a transistor switch. In some embodiments, the selected mode of communication includes at least one of a wireless connection according to a wireless communication standard and a wired connection. In some embodiments, the one or more electrical loads includes a rating of 40 amps or greater.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0013] The accompanying drawings, which are incorporated herein and form part of the specification, illustrate embodiments and, together with the description, further serve to explain the principles of the embodiments and to enable a person skilled in the relevant art(s) to make and use the embodiments.

[0014] FIG. 1 illustrates an electrical system according to an embodiment.

[0015] FIG. 2 illustrates an electrical system according to an embodiment.

[0016] FIG. 3 illustrates an electrical system according to an embodiment.

[0017] FIG. 4 illustrates a schematic diagram of a smart load panel in communication with a network according to an embodiment.

[0018] FIG. 5 illustrates a load breaker according to an embodiment.

[0019] FIG. 6 illustrates a load breaker according to an embodiment.

[0020] FIG. 7 illustrates a load breaker according to an embodiment.

[0021] FIG. 8 illustrates a block diagram showing aspects of a method of integrating a plurality of electrical loads to a backup side of an energy control system according to an embodiment.

[0022] FIG. 9 illustrates a chart showing modes of communication associated with a smart load panel according to an embodiment.

[0023] The features and advantages of the embodiments will become more apparent from the detail description set forth below when taken in conjunction with the drawings. A person of ordinary skill in the art will recognize that the drawings may use different reference numbers for identical, functionally similar, and/or structurally similar elements, and that different reference numbers do not necessarily indicate distinct embodiments or elements. Likewise, a person of ordinary skill in the art will recognize that functionalities described with respect to one element are equally applicable to functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

[0024] Embodiments of the present disclosure are described in detail with reference to embodiments thereof as illustrated in the accompanying drawings. References to “one

embodiment,” “an embodiment,” “some embodiments,” “certain embodiments,” etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0025] The term “about” or “substantially” or “approximately” as used herein refer to a considerable degree or extent. When used in conjunction with, for example, an event, circumstance, characteristic, or property, the term “about” or “substantially” or “approximately” can indicate a value of a given quantity that varies within, for example, 1-15% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 5\%$, $\pm 10\%$, or 15% of the value), such as accounting for typical tolerance levels or variability of the embodiments described herein.

[0026] The terms “upstream” and “downstream” as used herein refer to the location of a component of the electrical system with respect to the direction of current or power supply. For example, a first component is located “upstream” of a second component when current is being supplied from the first component to the second component, and a first component is located “downstream” of a second component when current is being supplied from the second component to the first component.

[0027] The terms “microgrid” and “backup mode” as used herein refer to group of interconnected loads (e.g., plurality of backup loads) and power distribution resources (e.g., backup PV power generation system, energy storage system, and energy control system) that function as a single controllable power network independent to the utility grid.

[0028] The following examples are illustrative, but not limiting, of the present embodiments. Other suitable modifications and adaptations of the variety of conditions and parameters normally encountered in the field, and which would be apparent to those skilled in the art, are within the spirit and scope of the disclosure.

[0029] Typical single backup power supply systems usually do not meet the load demands of particular loads (e.g., 40 A or greater, such as an air conditioner, an electric vehicle charger, a pool pump, heat pump, well pump, etc.) during backup mode due to the limited storage capacity, current, and power of the energy storage system. Accordingly, these large loads are typically installed on the non-backup side of the electrical system to avoid overload or reduce the load demand of the backup power supply system. Moving large loads to the non-backup side, however, typically requires specific types of breakers or subpanels, rendering the installation process cumbersome and expensive.

[0030] Thus, there is a need for new systems and procedures that allow large loads to be electrically coupled to the backup side of an electrical system without overloading or excessively draining backup power supply during the backup mode. Furthermore, there is a need for systems and procedures that allow users to selectively disconnect large loads from backup power supply during backup mode.

[0031] According to embodiments described herein, the electrical systems of the present disclosure can overcome one or more of these deficiencies, for example, by providing a smart load panel electrically coupled to an energy control system that is electrically coupled to a backup power source (e.g., PV system and/or energy storage system). The smart load panel is configured to maintain electrical connection of the plurality of backup loads to the energy control system when the energy control system is in the on-grid mode and to electrically disconnect one or more of the plurality of backup loads from the energy control system when the energy control system is in the backup mode. By disconnecting one or more (e.g., large) backup loads when the electrical system switches to the backup mode, the smart load panel can allow larger loads, such as, for example, an air conditioner, an over range, electrical vehicle, etc. to be electrically coupled to the backup side of

a microgrid interconnection device without posing a risk of AC overload or fast draining of the energy storage system during the backup mode.

[0032] FIGS. **1-3** show an energy control system **110** for controlling the operation of an electrical system **100** according to some embodiments. Electrical system **100** can include, for example, an energy storage system **150**, a backup photovoltaic (“PV”) power system **160**, a plurality of electrical loads **170**, a connection (e.g., a power bus with a subpanel and/or meter **182**) to a utility grid **180**, and/or a non-backup PV power generation system (e.g., non-backup PV power generation system **190** shown in FIG. **2**). In some embodiments, energy control system **110** can control the power distribution between energy storage system **150**, backup PV power generation system **160**, the plurality of electrical loads **170**, the connection to the utility grid **180**, and/or non-backup PV power generation system **190**. In some embodiments, energy control system **110** and electrical system **100** can include any component or be operated in any way, as disclosed in U.S. application Ser. No. 16/811,832, filed Mar. 6, 2020, titled “ENERGY CONTROL SYSTEM,” the entirety of which is incorporated herein by reference. In some embodiments, energy storage system **150**, backup PV power generation system **160**, and/or at least one of the electrical loads **170** (e.g., plurality of backup loads **172**) can be located on a backup side **104** of energy control system **110** such that energy control system **110**, energy storage system **150**, backup PV power generation system **160**, and/or at least one of the electrical loads **170** (e.g., plurality of backup loads **172**) can be configured as a single controllable power network independent of utility grid **180**. In some embodiments, utility grid **180** and/or non-backup PV power generation system **190** is electrically coupled to a non-backup side **106** of energy control system **110**.

[0033] In some embodiments, energy storage system **150** can include one or more batteries **152** configured to store electrical energy generated by backup PV power generation system **160**. In some embodiments, energy storage system **150** can include a storage converter **154** (e.g., inverter) electrically coupled to the batteries **152** by a direct current (DC) bus **153** and electrically coupled to energy control system **110** by an alternating current (AC) bus **140**. In some embodiments, storage converter **154** can be configured to convert the DC current discharged from batteries **152** to an AC current that emulates power characteristics (e.g., voltage magnitude and frequency) of utility grid **180**, such as for example, split phase AC at 240V/120V. In some embodiments, storage converter **154** can be configured to convert AC to DC. In some embodiments, storage converter **154** can be configured to adjust a charging rate and/or a discharging rate of the one or more batteries **152**. In some embodiments, storage converter **154** can be configured to adjust the frequency of AC power (e.g., AC voltage) supplied to backup side **104** (e.g., the frequency of microgrid) of energy control system **110**. In some embodiments, storage converter **154** can be configured to adjust the frequency of AC power supplied from energy storage system **150** to energy control system **110**. In some embodiments, storage converter **154** can be configured adjust the frequency of AC power supplied by backup PV power generation system **160**.

[0034] In some embodiments, storage converter **154** can include a controller **155** having a processor configured to process input signals and send commands via output signals. In some embodiments, controller **155** can include memory for storing, for example, information about energy storage system **150**, backup PV power generation system **160**, non-backup PV power generation system **190**, and/or energy control system **110**. In some embodiments, controller **155** can include firmware stored in the memory of controller **155** for controlling operation of storage converter **154** and/or battery **152**. In some embodiments, the firmware of controller **155** can include algorithms that enable the controller **155** to process electronic data received from energy control system **110** and/or backup PV power generation system **160**. In some embodiments, execution of the stored algorithms can allow controller **155** to detect frequency of AC power supplied by backup PV power generation system **160**, charging/discharging rate of batteries **152**, and/or state of charge of batteries **152**. In some embodiments, execution of the firmware can allow the controller **155** to adjust charging/discharging rate of batteries **152** and/or adjust frequency of AC power supplied to

the backup PV power generation system **160** based on the processed electronic data and/or detected measurements.

[0035] In some embodiments, backup PV power generation system **160** can include one or more power generation arrays (e.g., a photovoltaic panel array), and each power generation array can include one or more power generation units **162** (e.g., a photovoltaic panel) configured to generate electrical energy. In some embodiments, backup PV power generation system **160** can include one or more PV converters (e.g., a micro-inverter). In some embodiments, the PV converter can include any type of components (e.g., an inverter) such that the PV converter is configured to convert DC to AC or vice versa. In some embodiments, at least one PV converter can synchronize the phase of the power feed to split-phase AC that emulates power characteristics (e.g., voltage magnitude and frequency) of utility grid **180**, such as for example, split phase AC at 240V/120V. In some embodiments, the PV converter can be a part of power generation unit **162**. In some embodiments, one, two, three, four, or more power generation units can be interconnected to a single PV converter (e.g., a string inverter). In some embodiments, backup PV power generation system **160** can include one or more power optimizers such as, for example, DC power optimizers. In some embodiments, backup PV power generation system **160** can include a feed circuit configured to distribute power to the energy control system **110**.

[0036] In some embodiments, the plurality of electrical loads **170** can be separated into backup load(s) **172** and non-backup load(s) **174**. In some embodiments, a plurality of backup loads **172** include one or more essential loads that continue to receive power from the backup PV power generation system **160** and/or energy storage system **150** during a power grid outage, and a plurality of non-backup loads **174** includes one or more non-essential loads that do not receive power from the backup PV power generation system **160** and/or energy storage system **150** during a utility power outage. In the context of the present disclosure, an electrical load can be, for example, one or more devices or systems that consume electricity. In some embodiments, the plurality of electrical loads **170** can include all or some of the electrical devices associated with a building (e.g., a residential home). In some embodiments, the plurality of electrical loads **170** can include 240-volt loads. In some embodiments, the plurality of electrical loads **170** can include, for example, an electric range/oven, an air conditioner, a heater, a hot water system, a swimming pool pump, and/or a well pump. In some embodiments, the plurality of electrical loads **170** can include 120-volt loads. In some embodiments, the plurality of electrical loads **170** can include, for example, power outlets, lighting, networking and automation systems, a refrigerator, a garbage disposal unit, a dishwasher, a washing machine, other appliances, a septic pump, electric vehicle charger, and/or an irrigation system.

[0037] In some embodiments, non-backup PV power generation system **190** can include one or more power generation arrays (e.g., a photovoltaic panel array), and each power generation array can include one or more power generation units (e.g., a photovoltaic panel). In some embodiments, non-backup PV power generation system **190** can include one or more PV converters. In some embodiments, PV converter can include the features of any one of the converters described herein.

[0038] In some embodiments, energy control system **110** can include any number of interconnections to control the flow of energy between energy storage system **150**, backup PV power generation system **160**, the plurality of electrical loads **170**, utility grid **180**, and/or non-backup PV power generation system **190**. For example, in some embodiments, energy control system **110** can include a grid interconnection **184** electrically coupled to a utility grid **180** so that grid power is distributed to energy control system **110**. In some embodiments, grid interconnection **184** can include a main overcurrent protection device **183** that is electrically disposed between utility grid **180** and other components of energy control system **110**. In some embodiments, energy control system **110** can include a non-backup power bus **111** (e.g., 125 A rating bus) having one or more non-backup load interconnections **112** electrically coupled to the plurality of non-backup loads **174** and a non-backup PV interconnection **113** electrically coupled to non-backup PV power

generation system **190**. In some embodiments, energy control system **110** can include a backup power bus **114** (e.g., 200 A rating bus) having one or more backup load interconnections **115** electrically coupled to the plurality of backup loads **172** and a storage interconnection **116** electrically coupled to energy storage system **150**. In some embodiments, energy control system **110** can include a backup photovoltaic interconnection **117** (e.g., 125 A rating bus) electrically coupled to backup PV power generation system **160**. In the context of the present disclosure, an interconnection includes any suitable electrical structure, such as a power bus, wiring, a panel, etc., configured to establish electrical communication between two sets of circuits. Any one of interconnections **112**, **113**, **115**, **116**, **117**, and **184** can include an AC bus, a panel, a sub-panel, a circuit breaker, any type of conductor, or a combination thereof.

[0039] In some embodiments, energy control system **110** can include a microgrid interconnection device **120** (e.g., an automatic transfer or disconnect switch) electrically coupled to non-backup power bus **111** (e.g., located on a load side of microgrid interconnection device **120**) and backup power bus **114** (e.g., located on a line side of microgrid interconnection device **120**), such that microgrid interconnection device **120** is electrically coupled to non-backup load interconnection **112**, non-backup PV interconnection **113**, backup load interconnection **115**, storage interconnection **116**, and/or backup PV interconnection **117**. In some embodiments, microgrid interconnection device **120** is electrically coupled (e.g., directly) to grid interconnection **184**. In the context of the present disclosure, a microgrid interconnection device can be, for example, any device or system that is configured to automatically connect circuits, disconnect circuits, and/or switch one or more loads between power sources. In some embodiments, microgrid interconnection device **120** can include any combination of switches, relays, and/or circuits to selectively connect and disconnect respective interconnections **113**, **115**, **116**, **117**, and **184** electrically coupled to energy control system **110**. In some embodiments, such switches can be automatic disconnect switches that are configured to automatically connect circuits and/or disconnect circuits. In some embodiments, such switches can be transfer switches that are configured to automatically switch one or more loads between power sources.

[0040] In some embodiments, microgrid interconnection device **120** can be configured to operate in an on-grid mode (e.g., closed), in which microgrid interconnection device **120** electrically connects the backup power bus **114** to both the non-backup power bus **111** and grid interconnection **184**. In some embodiments, when operating in the on-grid mode, microgrid interconnection device **120** can be configured to distribute electrical energy received from utility grid **180** and/or non-backup PV power generation system **190** to backup loads **172**. In some embodiments, when operating in the on-grid mode, microgrid interconnection device **120** can be configured to distribute electrical energy received from energy storage system **150** and/or backup PV power generation system **160** to non-backup loads **174** and/or utility grid **180**.

[0041] In some embodiments, microgrid interconnection device **120** can be configured to operate in a backup mode, in which microgrid interconnection device **120** electrically disconnects both non-backup power bus **111** and grid interconnection **184** from backup power bus **114** and backup PV interconnection **117**. In some embodiments, when operating in the backup mode, microgrid interconnection device **120** can disrupt electrical connection from non-backup PV power generation system **190** from reaching backup loads **172**. In some embodiments, when operating in the backup mode, microgrid interconnection device **120** can disrupt electrical connection between backup loads **172** and utility grid **180**. In some embodiments, when operating in the backup mode, microgrid interconnection device **120** can disrupt electrical connection from energy storage system **150** and/or backup PV power generation system **160** to non-backup loads **174** and/or utility grid **180**.

[0042] In some embodiments, energy control system **110** can include a controller **122** in communication with microgrid interconnection device **120** and configured to control the distribution of electrical energy between energy storage system **150**, backup PV power generation

system **160**, the plurality of electrical loads **170**, utility grid **180**, and/or non-backup PV power generation system **190**. In some embodiments, controller **122** can be configured to detect the status (e.g., power outage or voltage restoration) of grid interconnection **184** and switch microgrid interconnection device **120** between the on-grid mode and the backup mode based on the status of grid interconnection **184**. If the status of grid interconnection **184** indicates a power outage, controller **122** can be configured to switch microgrid interconnection device **120** to the backup mode. If the status of grid interconnection **184** indicates a voltage restoration, controller **122** can be configured to switch microgrid interconnection device **120** to the on-grid mode.

[0043] In some embodiments, energy control system **110** includes a PV monitoring system **130**. In some embodiments, PV monitoring system **130** includes a communication interface (e.g., one or more antennas) for sending and/or receiving data over a wireless network. In some embodiments, energy control system **110** includes one or more load meters that monitor the current or voltage through certain elements of electrical system **100** and transmit data indicating the monitored current or voltage to PV monitoring system **130** and controller **122**. For example, a load meter can monitor the flow of power from microgrid interconnection device **120** to backup load interconnection **115**. A load meter can monitor the flow of electricity from microgrid interconnection device **120** to backup PV interconnection **117** and non-backup PV interconnection **113**. A load meter can monitor the flow of electricity from utility grid **180** to microgrid interconnection device **120**.

[0044] In some embodiments, PV monitoring system **130** can include a site consumption current transformer **132** (site CT) for monitoring the quantity of energy consumption by the plurality of electrical loads **170**. In some embodiments, site CT **132** can be operatively connected to grid interconnection **184**. In some embodiments, PV monitoring system **130** can include a PV production CT **134** for monitoring the quantity of PV energy outputted from backup PV power generation system **160**. In some embodiments, PV production CT **134** can be operatively linked to backup PV interconnection **117**.

[0045] In some embodiments, PV monitoring system **130** can read timeseries data and/or disable a reconnection timer of backup PV power generation system **160** and/or non-backup PV power generation system **190**. In some embodiments, PV monitoring system **130** can initiate a grid reconnection timer of backup PV power generation system **160**. In some embodiments, PV monitoring system **130** can communicate with a battery monitoring system (“BMS”) of energy storage system **150**. In some embodiments, PV monitoring system **130** can communicate with energy storage system **150** and can, for example, read timeseries data, read power information, write charge/discharge targets, and/or write “heartbeats.” In some embodiments, PV monitoring system **130** can receive status and/or power information from microgrid interconnection device **120**.

[0046] In some embodiments, controller **122** can be linked (e.g., wired or wirelessly) to PV monitoring system **130** such that controller **122** receives electronic data related to backup PV power generation system **160** and/or non-backup PV power generation system **190** from PV monitoring system **130**. In some embodiments, controller **122** can transmit commands to PV monitoring system **130** to adjust (e.g., increase or decrease) power output of backup PV power generation system **160** and/or non-backup PV power generation system **190** based on received data. In some embodiments, controller **122** can be configured as a master controller and PV monitoring system **130** can be configured to communicate electronic data (e.g., status of power generation) with controller **122** such that controller **122** controls control energy distribution based on the electronic data transmitted by PV monitoring system **130**.

[0047] In some embodiments, controller **122** can receive and transmit electronic data (e.g., computer-processable data and/or information represented by an analog or digital signal) over a network, such as, for example, Wireless Local Area Network (“WLAN”), Campus Area Network, Metropolitan Area Network, or Wide Area Network (“WAN”), with components of energy storage

system **150**, backup PV power generation system **160**, non-backup PV power generation system **190**, a user's device (e.g., user's smartphone or personal computer), smart device (e.g., load meter) and/or smart appliances (e.g., smart outlets, smart plugs, smart bulbs, smart washers, smart refrigerators). In some embodiments, electronic data can include timeseries data, alerts, metadata, outage reports, power consumption information, backup power output information, service codes, runtime data, etc.

[0048] In some embodiments, controller **122** can receive electronic data (e.g., from a load meter) related to load consumption of the plurality of electrical loads **170**, including backup loads **172** and/or non-backup loads **174**. In some embodiments, electronic data related to the plurality of electrical loads **170** can include information regarding the amount of power consumed by the plurality of electrical loads **170** (including backup loads **172** and/or non-backup loads **174**) and the times at which the power was consumed by the plurality of electrical loads **170**. In some embodiments, controller **122** can use the collected electronic data to determine a load average per circuit and/or a load average per smart device corresponding to discrete blocks of time throughout the day. For example, time blocks can be broken down into 1-hour blocks, 2-hour blocks, 3-hour blocks, or other time blocks, including, for example, user-designated time blocks (e.g., times when the user may be asleep, at home, or out of the house). In some embodiments, controller **122** can use the collected data to determine an energy demand based on the amount of power consumed by the plurality of electrical loads **170**.

[0049] In some embodiments, controller **122** can create a time-of-use library (e.g., a database or other structured set of data) that can define a circuit load average for each load and/or a smart device load average for each smart device with respect to the discrete blocks of time throughout the day. In some embodiments, controller **122** can use this information to determine which backup loads **172** receive power as a default during a grid power outage. In some embodiments, controller **122** can use this information to average load consumption by the plurality of backup loads **172** and/or non-backup loads **174** profiled over a day of time.

[0050] In some embodiments, the converter of backup PV power generation system **160** can transmit to controller **122** electronic data related to backup PV power generation system **160**. In some embodiments, electronic data related to backup PV power generation system **160** can include a current (e.g., an instantaneous) power output of backup PV power generation system **160**. In some embodiments, electronic data related to backup PV power generation system **160** can include historical power output measurements of backup PV power generation system **160** recorded over an extended period of time (e.g., days, weeks, months). In some embodiments, electronic data related to backup PV power generation system **160** can include the average power output of the backup PV power generation system **160**, for example, profiled over a day. In some embodiments, controller **122** can calculate a predicted power output of backup PV power generation system **160** based on the historical data and other information, such as, for example, weather forecasts and state of the power generation arrays (e.g., power output capacity). In some embodiments, controller **122** uses the average power output of the backup PV power generation system **160** as a predicted power output for controlling operations of electrical system **100**.

[0051] In some embodiments, storage converter **154** of energy storage system **150** can transmit to controller **122** electronic data related to energy storage system **150**. In some embodiments, electronic data related to energy storage system **150** can include information relating to the amount of energy currently stored in energy storage system **150** (e.g., a current state of charge) and/or the amount of energy that energy storage system **150** is capable of absorbing (e.g., via charging). In some embodiments, electronic data related to energy storage system **150** can include the amount of energy being discharged (e.g., current discharging rate and/or the duration of the battery discharging) or predicted to be discharged (e.g., based on a time-of-use library) from energy storage system **150**.

[0052] In some embodiments, electrical components (e.g., interconnections, switches, relays, AC

bus) of energy control system **110** can be integrated into a single housing. For example, as shown in FIG. 2, in some embodiments, energy control system **110** can include a housing **102**. In some embodiments, electrical components (e.g., interconnections, switches, relays, AC bus) of energy control system **110** can be disposed in multiple housings, such as for, example, a panel disposed in a home building and a subpanel disposed in a garage or pool house.

[0053] As shown in FIG. 2, in some embodiments, electrical system **100** can include a main service panel **200** integrated with utility meter **182**. In some embodiments, main service panel **200** can include a main circuit breaker **204**. In some embodiments, main service panel **200** can be connected to one or more electrical loads **210**. In some embodiments, electrical system **100** can include a subpanel **220** located downstream of main service panel **200** connected to a plurality of electrical loads **230**. In some embodiments, the plurality of loads **230** can include small loads having a breaker size of 40 A or less (e.g., lighting, router, television) and large loads having a breaker size greater than 40 A (e.g., air conditioner system, oven).

[0054] In some embodiments, as shown in FIGS. 2 and 3, electrical system **100** can include a smart load panel **300** electrically coupled to backup side **104** of energy control system **110**. In some embodiments, the plurality of electrical loads **210** can be decoupled from the main circuit breaker **204** and electrically coupled to smart load panel **300**. In some embodiments, smart load panel **300** can be electrically coupled to subpanel **220** and/or directly to the plurality of electrical loads **230**. In some embodiments, subpanel **220** can be implemented as smart load panel **300** by replacing or supplementing subpanel **220** with circuitry components that allow a controller to selectively disconnect electrical loads **230** from energy control system **110**. In some embodiments, smart load panel **300** can be electrically coupled in series between the load appliance wiring of electrical loads **230** and a load breaker of a panel or subpanel. In some embodiments, smart load panel **300** can selectively disconnect each of the plurality of electrical loads **210**, subpanel **220**, and/or electrical loads **230** from energy control system **110** so that power supplied from PV power generation system **160** and/or energy storage system **150** is not distributed to the selected electrical load. In some embodiments, smart load panel **300** can selectively disconnect one or more of the plurality of electrical loads **210**, subpanel **220**, and/or electrical loads **230** from energy control system **110** when energy control system **110** is in the on-grid mode and/or when energy control system **110** is in the backup mode. In some embodiments, smart load panel **300** can selectively disconnect a majority of the plurality of electrical loads **210**, subpanel **220**, and/or electrical loads **230** from energy control system **110** when energy control system **110** is in the on-grid mode and/or when energy control system **110** is in the backup mode. In some embodiments, smart load panel **300** can selectively disconnect each of the plurality of electrical loads **210**, subpanel **220**, and/or electrical loads **230** from energy control system **110** when energy control system **110** is in the on-grid mode and/or when energy control system **110** is in the backup mode.

[0055] In some embodiments, smart load panel **300** can be configured to maintain electrical connection of electrical loads **210**, subpanel **220**, and/or electrical loads **230** to energy control system **110** when the energy control system **110** is in the on-grid mode. In some embodiments, smart load panel **300** can be configured to electrically disconnect any one of electrical loads **210**, subpanel **220**, and/or electrical loads **230** from energy control system **110** when the energy control system **110** switches to the backup mode. In some embodiments, the selected load is a large load intended to be disconnected from the energy control system **110** during the backup mode to prevent unwanted power drainage of the energy storage system **150**. In the context of present disclosure, a large electrical load as used herein refers to a component configured to receive power from 40 A current source or greater and/or a component configured to consume 2000 or more watts. By selectively disconnecting any one of electrical loads **210**, subpanel **220**, and/or electrical loads **230** from energy control system **110**, smart load panel **300** can allow all the loads of electrical system **100** to be electrically coupled to backup side **104** of energy control system **110**, without having any electrical loads coupled to the non-backup side **106** of energy control system **110**. Accordingly,

smart load panel **300** can save users cost and simplify installation of energy control system **110** by not having to move large loads to the non-backup side **106** of the energy control system **110**. [0056] In some embodiments, smart load panel **300** can be configured as an external service panel integrated with backup side **104** of energy control system **110**. With reference to FIG. 4, in some embodiments, smart load panel **300** can include an input port **310** for receiving power (e.g., AC voltage) from energy control system **110**. In some embodiments, smart load panel **300** can include one or more load breakers **320** electrically coupled to input port **310** and a respective electrical load (e.g., via a branch power line having a 60 A rating). For example, in some embodiments, smart load panel **300** can include eight load breakers **320** to electrically connect smart load panel **300** to at least eight branch circuits that are electrically coupled to any one of electrical loads **210**, subpanel **220**, and/or electrical loads **230**. In some embodiments, smart load panel **300** can include one or more bus bars **312** (e.g., 300 V rating bus bar) coupled to input port **310**, and the one or more load breakers **320** are coupled to a respective bus bar **312**. In some embodiments, smart load panel **300** can include any other components (e.g., a printed circuit board) suitable for connecting load breaker **320** to input port **310**. Through the one or more load breakers **320**, power received at input port **310** can be distributed dynamically to the plurality of electrical loads **210**, subpanel **220**, and/or electrical loads **230**.

[0057] In some embodiments, smart load panel **300** can include a load panel controller **400** configured to actuate the one or more load breakers **320** to selectively disconnect any one of electrical loads **210**, subpanel **220**, and/or electrical loads **230** from energy control system **110**. In some embodiments, load panel controller **400** can actuate the one or more load breakers **320** to disrupt the electrical connection to its respective electrical load within a predetermined time period (e.g., a response time in a range between approximately 10 milliseconds and approximately 40 milliseconds) that is compliant with state or national codes and product standards. In some embodiments, load panel controller **400** can be configured to: (1) selectively disconnect large loads (e.g., rated at 40 A or greater) from energy control system **110** when energy control system **110** switches from on-grid mode to backup mode; (2) selectively disconnect predetermined electrical loads from energy control system **110** when energy control system **110** switches from on-grid mode to backup mode; and/or (3) selectively disconnect any one backup loads **172** and electrical loads **210**, **230** from energy control system **110** when energy control system **110** switches from on-grid mode to backup mode. In some embodiments, load panel controller **400** can include microprocessor, dedicated or general purpose circuitry (e.g., an application-specific integrated circuit and/or a field-programmable gate array), a suitably programmed computing device, and/or any other suitable circuitry for controlling the operation of the one or more load breakers **320**.

[0058] In some embodiments, smart load panel **300** can include an enclosure **302** housing input port **310**, the one or more load breakers **320**, and load panel controller **400**. In some embodiments, load panel controller **400** can be disposed its own housing separated from enclosure **302** of smart load panel **300**. In some embodiments, enclosure **302** can be configured for indoor (e.g., NEMA-1 rated enclosure) and/or outdoor (e.g., NEMA-3 rated enclosure).

[0059] In some embodiments, the one or more load breakers **320** can include any type of circuitry component suitable for disrupting an electrical connection between the respective electrical load (e.g., at least one of electrical loads **210**, subpanel **220**, and/or electrical loads **230**) and energy control system **110**.

[0060] In some embodiments, as shown in FIG. 5, for example, load breaker **320** can be configured as an electromechanical circuit breaker (e.g., a switch device having a coil, an armature, and contactors). In some embodiments, load breaker **320** can include a line side **502** electrically coupled to energy control system **110** (e.g., via input port **310**), and a load side **504** electrically coupled to a respective electrical load (e.g., any one of electrical loads **210** and/or electrical loads **230**). In some embodiments, load breaker **320** can include a switch **510** disposed along a line conductor **506** that receives AC voltage and current from line side **502** and distributes voltage and

current to load side **504**. In some embodiments, switch **510** can be configured to move between a closed position, in which current is allowed to flow from line side **502** to load side **504**, and an open position, in which current is disrupted between line side **502** and load side **504**.

[0061] In some embodiments, load breaker **320** can include sensor circuitry **520** (e.g., a standard resistor chain and signal filter) configured to measure voltage, current, and/or frequency across the line conductor **506**. In some embodiments, sensor circuitry **520** can include any type of circuitry component (e.g., a voltmeter, resistor chain, signal filter, a potential transformer, and/or a current transformer), to measure voltage, current, and/or frequency across line conductor **506**. In some embodiments, sensor circuitry **520** can be electrically coupled to first phase line, second phase line, neutral line, and/or ground line of the circuit coupled to the respective electrical load (e.g., any one of electrical loads **210** and/or electrical loads **230**) to measure voltage between first phase line, second phase line, and/or neutral line.

[0062] In some embodiments, load panel controller **400** can be configured to receive measurements (e.g., voltage level, current, and/or frequency) from sensor circuitry **520** of the one or more load breakers **320**. In some embodiments, load panel controller **400** can be configured to transmit drive signals directly to switch **510**. In some embodiments, load panel controller **400** can be configured to transmit drive signals to an actuator (e.g., solenoid, motor) to move the switch **510** between the open and closed positions. In some embodiments, load panel controller **400** can be configured to transmit a drive signal through a driver to switch **510** and/or actuator of switch **510** to actuate movement between the open and closed positions. In some embodiments, load panel controller **400** can include an analog-to-digital converter to convert analog signals received from sensor circuitry **520** to digital signals. In some embodiments, load panel controller **400** can include a processor for processing input signals and generating the drive signals.

[0063] In some embodiments, as shown in FIG. 6, for example, load breaker **320** can be configured as a smart relay (e.g., a switch device having a coil, an armature, and contactors). Similar to the embodiment shown in FIG. 5, load breaker **320** shown in FIG. 6 can include line side **502**, load side **504**, line conductor **506**, and sensor circuitry **520**. In some embodiments, load breaker **320** can include a relay **530**, instead of a switch, electrically coupled to line conductor **506**. In some embodiments, relay **530** can be configured to switch between a closed position to permit electrical connection between line side **502** and load side **504** and an open position to disrupt electrical connection between line side **502** and load side **504**. In some embodiments, load panel controller **400** can be configured to transmit a drive signal to relay **530** to actuate relay **530** to switch between open and/or closed positions.

[0064] In some embodiments, as shown in FIG. 7, for example, load breaker **320** can be configured as a solid-state switch (e.g., semiconductor device having a transistor or integrated-circuit). Similar to the embodiment shown in FIG. 5, load breaker **320** shown in FIG. 7 can include line side **502**, load side **504**, line conductor **506**, and sensor circuitry **520**. In some embodiments, load breaker **320** can include a transistor **540**, instead of a relay, electrically coupled to line conductor **506**. In some embodiments, transistor **540** can be configured to switch between a closed setting to permit electrical connection between line side **502** and load side **504** and an open setting to disrupt electrical connection between line side **502** and load side **504**. In some embodiments, load panel controller **400** can be configured to transmit a drive signal to a gate of transistor **540** to actuate transistor **540** to switch between open and/or closed settings.

[0065] In some embodiments, smart load panel **300** can be configured to detect an electrical characteristic (e.g., voltage, current, and/or frequency) of the power supplied from energy control system **110** to the plurality of electrical loads **210**, subpanel **220**, and/or electrical loads **230**. In some embodiments, smart load panel **300** can use sensor circuitry **520** of load breaker **320** to detect electrical characteristics of the power supply. In some embodiments, smart load panel **300** can include additional sensor circuitry (e.g., a voltmeter, resistor chain, signal filter, a potential transformer, and/or a current transformer) located at input port **310** to measure voltage, current,

and/or frequency of power supplied from backup side **104** of energy control system **110**.

[0066] In some embodiments, load panel controller **400** can receive commands from another controller, such as, for example, controller **122** and/or PV monitoring system **130** of energy control system **110**, to control operation of the one or more load breakers **320**. For example, load panel controller **400** can communicate with another controller (e.g., controller **122** and/or PV monitoring system **130**) over a wired connection and/or wireless connection (e.g., via a Personal Area Network, WLAN, and/or WAN). In some embodiments, the wired connection for load panel controller **400** of smart load panel **300** can be implemented using Uni-logic signal or Multi-logic signal software, an Ethernet line, a Controller Area Network (CAN) bus, RS-232 cable and/or RS-485 cable. In some embodiments, load panel controller **400** can communicate with another controller (e.g., controller **122**, PV monitoring system **130**, and/or a user device **600**) according to a wireless communication protocol, such as wireless fidelity (Wi-Fi under IEEE 802.11), Bluetooth (under IEEE 802.15.1), Zigbee (under IEEE 802.15.4), an AC Power Line Communication (PLC), and/or a broadband cellular network (2G, 3G, 4G, and/or 5G networks).

[0067] In some embodiments, controller **122** and/or PV monitoring system **130** of energy control system **110** can monitor electronic data of electrical system **100** and transmit a command to load panel controller **400** of smart load panel **300** to open and/or close the one or more load breakers **320** based on monitored electronic data. In some embodiments, controller **122** or PV monitoring system **130** can process the electronic data within a set time period (e.g., about 5 milliseconds to about 10 milliseconds) and transmit a disconnection command to load panel controller **400** to actuate load breaker **320** within a set response period (e.g., 10 milliseconds to 40 milliseconds). Upon receiving a disconnection command from controller **122** or PV monitoring system **130**, load panel controller **400** can electrically disconnect any one of the plurality of electrical loads **210**, subpanel **220**, and the plurality of electrical loads **230** from energy control system **110**.

[0068] In some embodiments, as shown in FIG. 4, for example, load panel controller **400** can communicate with a user device **600** over a wireless network **350**. User device **600** can be, for example, a cell phone, smartphone, tablet computer, laptop computer, desktop computer, personal computer, wearable computer, smartwatch, or other computing device capable of connecting to a wireless network. In some embodiments, user device **600** can include a user interface **610**, such as a touch screen display for receiving user input and displaying information to a user. In some embodiments, user interface **610** can include a combination of touch screens, electromechanical buttons, and/or visual displays.

[0069] In some embodiments, user interface **610** of the user device **600** can display information about, for example, the type(s) of loads in electrical system **100**, groupings of loads in electrical system **100**, present and/or average power consumption of loads (e.g., an energy demand associated with the plurality of loads), the state of load breakers **320**, the amount of power available in energy storage system **150**, the amount of power produced by power generation system **160**, and/or other information relating to electrical system **100**. In some embodiments, user interface **610** of the user device **600** can display information about, for example, customizable name of the loads and/or the amount of energy consumed by a circuit over a customizable time period, such as, for example, a time period from approximately one minute to approximately six months (e.g., one day, one week, or one month). In some embodiments, a user can control load breakers **320** remotely via the user device **600**. For example, the user device **600** can be configured to transmit a disconnection command over network **350** to load panel controller **400**. Upon receiving a disconnection command from user device **600**, load panel controller **400** is configured to electrically disconnect any one of the plurality of electrical loads **210**, subpanel **220**, and/or electrical loads **230** from energy control system **110**.

[0070] In some embodiments, load panel controller **400** can control operation of the one or more load breakers **320** independently, without communicating with another controller. For example, smart load panel **300** can be configured to automatically disrupt the electrical disconnection

between a respective electrical load and energy control system **110** based on processing of a detected electrical characteristic, so that any one of the plurality of electrical loads **210**, subpanel **220**, and/or electrical loads **230** does not overload or rapidly drain the energy storage system **150**. In some embodiments, load panel controller **400** can monitor the frequency of the AC voltage transmitted to a respective load and determine whether energy control system **110** has switched from the on-grid mode to the backup mode based on the detected frequency of power supplied to the respective load. In some embodiments, load panel controller **400** can disconnect any one of the plurality of electrical loads **210**, subpanel **220**, and/or electrical loads **230** when monitored frequency exceeds a frequency deviation threshold (e.g., in a range from approximately 0.1 Hz to approximately 5 Hz, such as, for example, 0.5 Hz).

[0071] FIG. **8** shows an example block diagram illustrating aspects of a method **700** for integrating a plurality of electrical loads (e.g., the plurality of electrical loads **210**, subpanel **220**, and/or the plurality off electrical loads **230**) into electrical system **100** (e.g., a residential electrical system).

[0072] In some embodiments, method **700** can include a step **710** of determining a location of one or more of the plurality of electrical loads (e.g., the plurality of electrical loads **210**, subpanel **220**, and/or the plurality off electrical loads **230**) with respect to energy control system **110**. In some embodiments, step **710** can include determining the separation distance between the one or more electrical loads and energy control system **110**. In some embodiments, step **710** can include determining whether the electrical load is located on non-backup side **106** or backup side **104** of energy control system **110**. For example, the electrical load can be electrically coupled to a main service panel (e.g., service panel **200**) that is electrically coupled to non-backup side **106** of energy control system **110**. In another example, the electrical load is electrically coupled to a subpanel (e.g., subpanel **220**) electrically coupled to backup side **104** of energy control system **110**.

[0073] In some embodiments, method **700** can include a step **720** of determining a load control protocol associated with the one or more electrical loads (e.g., the plurality of electrical loads **210**, subpanel **220**, and/or the plurality off electrical loads **230**). In some embodiments, the load control protocol includes allowing a user to selectively connect and disconnect the one or more electrical loads on demand (e.g., via user device **600**). In some embodiments, the load control protocol includes automatically disconnecting the one or more electrical loads from energy control system **110** based on a detected frequency and/or voltage of AC power supply. In some embodiments, the load control protocol includes receiving commands from controller **122** and/or PV monitoring system **130** of energy control system **110** to selectively connect or disconnect one or more load breakers **320**. In some embodiments, the load control protocol includes allowing a controller (e.g., load panel controller **400**, controller **122**, PV monitoring system **130**, and/or user device **600**) to monitor the load consumption by the respective electrical load. For example, a controller (e.g., load panel controller **400**, controller **122**, PV monitoring system **130**, and/or user device **600**) can actively monitor power usage of the one or more electrical loads, and can turn off certain circuits and/or smart devices if power consumption (e.g., load demand associated with respective electrical loads **210**, subpanel **220**, and/or the plurality off electrical loads **230**) is too high (e.g., above a predetermined or pre-set threshold). In some embodiments, the load control protocol includes providing load control customization such that a series of defaults, rules, and/or checks are executed to determine which electrical loads are deactivated when energy control system **110** switches from the on-grid mode to the backup mode.

[0074] In some embodiments, method **700** can include a step **730** of selecting an electrical disconnection component (e.g., load breaker **320**) based on the determined load control protocol of the one or more electrical loads (e.g., the plurality of electrical loads **210**, subpanel **220**, and/or the plurality off electrical loads **230**). In some embodiments, the electrical disconnection component can be smart load panel **300** having the one or more load breakers **320**. In some embodiments, load breaker **320** can include an electromechanical breaker, a relay, and/or a transistor switch, as described herein.

[0075] In some embodiments, method **700** can include a step **740** of connecting the selected electrical disconnection component (e.g., smart load panel **300** with load breaker **320**) to the one or more electrical loads (e.g., the plurality of electrical loads **210**, subpanel **220**, and/or the plurality of electrical loads **230**) and connecting the electrical disconnection component to backup side **104** of the energy control system **110**. For example, in some embodiments, step **740** can include electrically decoupling the plurality of electrical loads **210** from main service panel **200** and electrically coupling the plurality of electrical loads **210** to one or more load breakers **320** of smart load panel **300**. In some embodiments, step **740** can include electrically coupling subpanel **220** to a respective load breaker **320** of smart load panel **300**. In some embodiments, step **740** can include implementing subpanel **220** as a smart load panel by installing remotely-controlled load breakers that are electrically coupled to the plurality of electrical loads **230**.

[0076] In some embodiments, method **700** can include a step **750** of selecting a mode of communication between the selected electrical disconnection component (e.g., smart load panel **300** with load breaker **320**) and energy control system **110** based on the determined location of the one or more electrical loads (e.g., the plurality of electrical loads **210**, subpanel **220**, and/or the plurality of electrical loads **230**) and the determined load control protocol. In some embodiments, the selected mode of communication in step **750** can include a wireless connection according to a wireless communication standard, such as, for example, wireless fidelity (Wi-Fi under IEEE 802.11), Bluetooth (under IEEE 802.15.1), Zigbee (under IEEE 802.15.4), an AC PLC, and/or a broadband cellular network (2G, 3G, 4G, and/or 5G networks). In some embodiments, the selected mode of communication in step **750** can include a wired connection, such as, for example, Uni-logic signal or Multi-logic signal software, an Ethernet line, a CAN bus, RS-232 cable, and/or RS-485 cable.

[0077] In some embodiments, the selected mode of communication in step **750** can account for the location of the one or more electrical loads relative to energy control system **110**. For example, if the one or more electrical loads are located within a predetermined range of energy control system **110** (e.g., within approximately 30 feet), the selected mode of communication between smart load panel **300** and energy control system **110** can be a wired connection and/or a Bluetooth connection. If the one or more electrical loads are located beyond a predetermined range of energy (e.g., more than approximately 30 feet), the selected mode of communication can be, for example, over a network according to a Wi-Fi protocol.

[0078] In some embodiments, the selected mode of communication in step **750** can account for the selected load control protocol. For example, if the selected load control protocol is automatically disconnecting the one or more electrical loads from energy control system **110** based on a detected frequency and/or voltage of AC power supply, the selected mode of communication can be a wired connection implemented by Uni-logic signal or Multi-logic signal software. If the selected load control protocol is allowing a user to selectively connect and disconnect the one or more electrical loads on demand, the selected mode of communication can be a wireless connection, such as Bluetooth, Wi-Fi, and/or cellular or AC PLC. In some embodiments, if the load control protocol is load monitoring and/or receiving commands from energy control system **110**, the selected mode of communication can be a wired connection, such as Ethernet line, CAN, and/or RS-485 cable, and/or a wireless connection, such as AC PLC, Bluetooth, and/or Wi-Fi.

[0079] In some embodiments, the selected mode of communication in step **750** can account for the feasibility of installing the communication connection with the electrical disconnection component (e.g., smart load panel **300** with one or more load breakers **320**). For example, wired connections, such as Ethernet line and CAN, can be more difficult to install, compared to implementing wireless connections, such as Wi-Fi and Bluetooth.

[0080] It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections can set forth one or more but not all exemplary embodiments of the present embodiments

as contemplated by the inventor(s), and thus, are not intended to limit the present embodiments and the appended claims in any way.

[0081] The present disclosure has been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

[0082] The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

Claims

1. An electrical system comprising: an energy control system; a utility grid electrically connected to the energy control system; an energy storage system electrically coupled to the energy control system; a photovoltaic (PV) generation system electrically connected to the energy storage system; and a plurality of loads electrically connected to the energy control system, wherein the energy control system is configured to electrically connect at least one of the utility grid, the energy storage system, or the PV generation system to the plurality of loads.
